Fourier series

Preliminaries

Periodic functions:

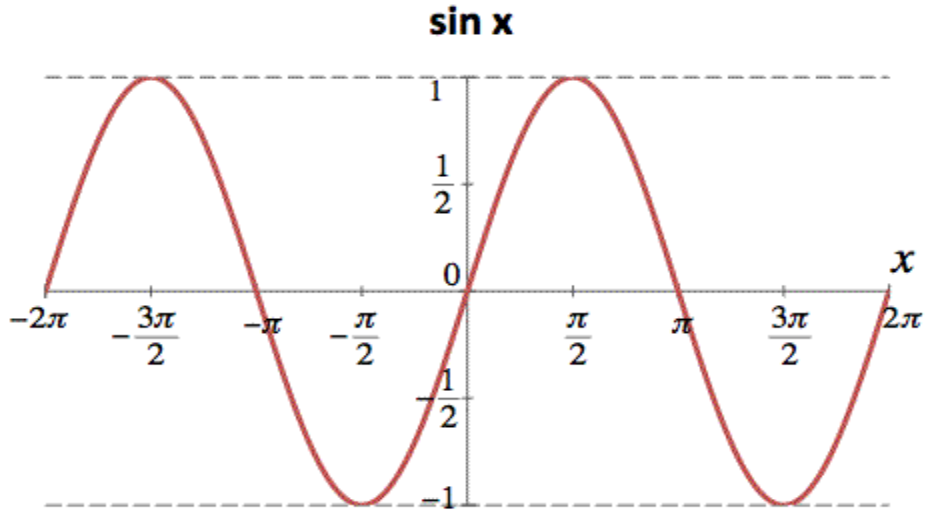
A function $f(t)$ is said to be periodic with period if there exists a positive constant T such that $f(t+T)=f(t)$ for all t.

Note: If $f(t)$ is period with period T, then $f(t+nT)=f(t)$ for all integers n.

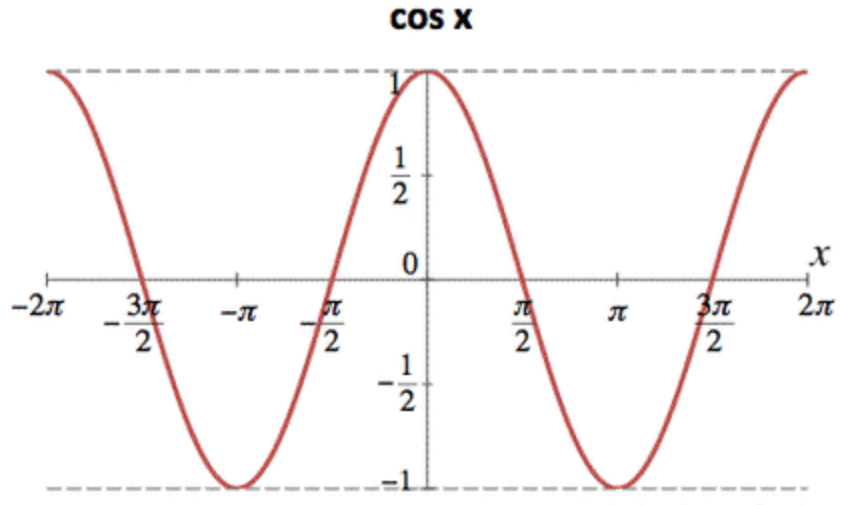
The smallest positive number T satisfying this property is called the primitive period or simply the period of the function $f(t)$.

The graph of a period function $f(t)$ with period T periodically repeats in an interval of width T. Hence it is sufficient to study the properties (nature) of the function in an interval of length T, in particular in the interval $[0,T]$, which is called one period of the function.

Example (1): The trigonometric functions $f(x)=\sin x$ and $f(x)=\cos x$ are periodic with period $T=2\pi$.



Graph of $f(x)=\sin x$



Graph of $f(x) = \cos x$

Example (2): The functions $\sin kt$ and $\cos kt$ have period $T = \dfrac{2\pi}{k}$, since

$$f\left(t + \frac{2\pi}{k}\right) = \sin\left(k(t + \frac{2\pi}{k}\right) = \sin(kt + 2\pi) = \sin kt.$$

Let $f(t)$ be a periodic function of t with the period T. Define a new variable x as follows.

$$\frac{x}{2\pi} = \frac{t}{T} \text{ or } x = \frac{2\pi}{T}t \text{ i.e., } t = \frac{T}{2\pi}x$$

Then $f(t) = f\left(\dfrac{T}{2\pi}x\right) = g(x)$ is a function of x.

Consider $g(x + 2\pi) = f\left(\dfrac{T}{2\pi}(x + 2\pi)\right) = f\left(\dfrac{T}{2\pi}x + T\right) = f(t + T) = f(t)$

$$= f\left(\frac{T}{2\pi}x\right) = g(x)$$

Therefore $g(x)$ is a periodic function of x with period 2π.

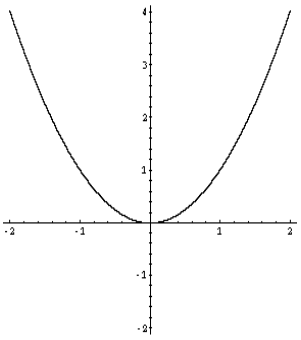
Note: If $f(x)$ and $g(x)$ are periodic functions of x with period T_1 and T_2 respectively, then $c_1 f(x) + c_2 g(x)$ is also periodic with period $T = lcm(T_1, T_2)$.

Even an Odd functions:

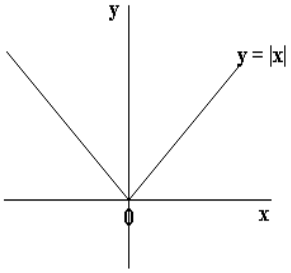
A function $f(t)$ is said to be even if $f(-t) = f(t)$ and odd if $f(-t) = -f(t)$.

Note that the graph of an even function is symmetric about $y-$axis, whereas the graph of an odd function is symmetric about the origin.

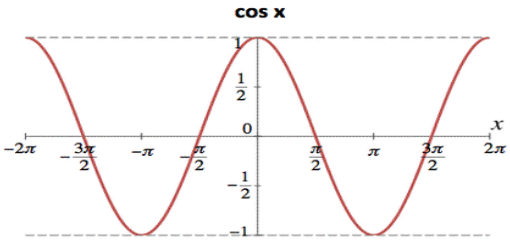
Exmple (1): $f(x) = x^2$, $f(x) = |x|$ and $f(x) = \cos x$ are all even functions of x.



Graph of $f(x) = x^2$

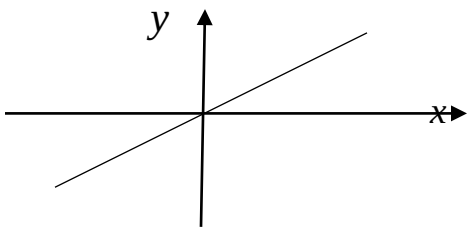


Graph of $f(x) = |x|$

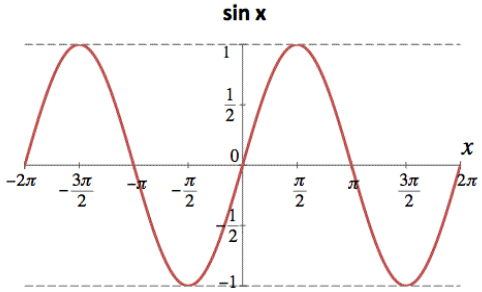


Graph of $f(x) = \cos x$

Example (2): $f(x) = x$, $\sin x$, $f(x) = x^3$, $f(x) = \sin x$, $f(x) = \tan x$ are all odd functions of x.



Graph of $f(x) = x$



Graph of $f(x) = \sin x$

Products of two even and two odd functions are even and the product of an even function with an odd function is odd.

Also, $\int_{-a}^{a} f(t)dt = \begin{cases} 2\int_{0}^{a} f(t)dt, & \text{if } f(t) \text{ is even} \\ 0, & \text{if } f(t) \text{ is odd} \end{cases}$.

Generalized formula for integration by parts

If u and v are functions of x, then

$$\int uvdx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \ldots$$

where $u' = \frac{du}{dx}$, $u'' = \frac{du'}{dx}$, $u''' = \frac{du''}{dx}, \ldots$ and

$$v_1 = \int vdx, \quad v_2 = \int v_1 dx, \quad v_3 = \int v_2 dx, \ldots$$

Example: $\int x^2 \sin 2x \, dx = x^2\left(\frac{-\cos 2x}{2}\right) - 2x\left(\frac{-\sin 2x}{4}\right) + 2\left(\frac{\cos 2x}{8}\right)$

Orthogonality of trigonometric functions:

$$\int_{c}^{c+T} \cos\frac{2n\pi}{T}t \, dt = \begin{cases} 0, & n \neq 0 \\ T, & n = 0 \end{cases}$$

$$\int_{c}^{c+T} \sin\frac{2n\pi}{T}t \, dt = 0 \text{ for all } n$$

$$\int_{c}^{c+T} \cos\frac{2m\pi}{T}t \, \sin\frac{2n\pi}{T}t \, dt = 0 \text{ for all } m \text{ and } n$$

$$\int_{c}^{c+T} \cos\frac{2m\pi}{T}t \ \cos\frac{2n\pi}{T}t \ dt = \begin{cases} 0, & \text{m} \neq n \\ \dfrac{T}{2}, & \text{m} = n \end{cases}$$

$$\int_{c}^{c+T} \sin\frac{2m\pi}{T}t \ \sin\frac{2n\pi}{T}t \ dt = \begin{cases} 0, & \text{m} \neq n \\ \dfrac{T}{2}, & \text{m} = n \end{cases}$$

Also,

$\cos n\pi = (-1)^n$, for all n , $\sin n\pi = 0$, for all n

$$\cos\frac{n\pi}{2} = \begin{cases} 0, & n \text{ odd} \\ (-1)^{\frac{n}{2}}, & n \text{ even} \end{cases}, \quad \sin\frac{n\pi}{2} = \begin{cases} 0, & n \text{ even} \\ (-1)^{\frac{n-1}{2}}, & n \text{ odd} \end{cases}$$

Definition: A function $f(t)$ is said to be piecewise continuous in an interval $[a, b]$ if it is discontinuous at finite number of points in the interval and wherever it is discontinuous, it has finite left and right hand limits.

Fourier series:

In solving many boundary valued problems involving ordinary and partial differential equations it is required to represent some functions as a sum of trigonometric functions cosine and sine. Such a series representation of a function $f(t)$ (which may be discontinuous); if exists, is called the trigonometric series expansion or the Fourier series expansion of $f(t)$.

The Fourier series is named in honour of Jean-Baptiste Joseph Fourier (1768–1830), who made important contributions to the study of trigonometric series, after preliminary investigations by Leonhard Euler, Jean le Rond d'Alembert, and Daniel Bernoulli. Fourier introduced the series for the purpose of solving the heat equation in a metal plate, publishing his initial results in his 1807 *Mémoire sur la propagation de la chaleur dans les corps solides* (*Treatise on the propagation of heat in solid bodies*), and publishing his *Théorie analytique de la chaleur* (*Analytical theory of heat*) in 1822. The *Mémoire* introduced Fourier analysis, specifically Fourier series. Through Fourier's research the fact was established that

an arbitrary (continuous) function can be represented by a trigonometric series. The first announcement of this great discovery was made by Fourier in 1807, before the French Academy.[3] Early ideas of decomposing a periodic function into the sum of simple oscillating functions date back to the 3rd century BC, when ancient astronomers proposed an empiric model of planetary motions, based on deferents and epicycles.

Fourier introduced such an expansion of periodic functions in terms of sine and cosine functions and hence it is called a Fourier series expansion. Many functions including some discontinuous periodic functions can be expanded in a Fourier series and hence are, in certain sense more universal than Taylor series expansions, which cannot be established for discontinuous functions. Fourier series solution method is a powerful tool in solving some ordinary and partial differential equations given with the initial or boundary conditions.

Although the original motivation was to solve the heat equation, it later became obvious that the same techniques could be applied to a wide array of mathematical and physical problems, and especially those involving linear differential equations with constant coefficients, for which the eigen solutions are sinusoids. The Fourier series has many such applications in electrical engineering, vibration analysis, acoustics, optics, signal processing, image processing, quantum mechanics, econometrics, thin-walled shell theory, etc.

A Fourier series is an expansion of a periodic function $f(x)$ in terms of an infinite sum of sines and cosines. Fourier series make use of the orthogonality relationships of the sine and cosine functions. The computation and study of Fourier series is known as harmonic analysis and is extremely useful as a way to break up an *arbitrary* periodic function into a set of simple terms that can be plugged in, solved individually, and then recombined to obtain the solution to the original problem or an approximation to it to whatever accuracy is desired or practical.

Definition:

Let $f(t)$ be a periodic function of t with period T, and is defined in an interval $[c, c+T]$. Then the expansion of the form

$f(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{2n\pi}{T}t\right) + b_n \sin\left(\frac{2n\pi}{T}t\right)\right)$, if exists called the Fourier series or Fourier expansion of $f(t)$. Here a_0, a_n and b_n are called Fourier coefficients.

Euler's formulae:

Given a periodic function $f(t)$ with period T, represented in $[c,c+T]$ by a Fourier series,

$f(t)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos\left(\frac{2n\pi}{T}t\right)+b_n\sin\left(\frac{2n\pi}{T}t\right)\right)$ ----------(1), to determine the coefficients a_0, a_n and b_n, we proceed as follows,

Integrating (1) we get,

$$\int_c^{c+T} f(t)dt=\int_c^{c+T}\left(a_0+\sum_{n=1}^{\infty}a_n\cos\left(\frac{2n\pi}{T}t\right)+b_n\sin\left(\frac{2n\pi}{T}t\right)\right)dt$$

$$=\int_c^{c+T}a_0dt+\sum_{n=1}^{\infty}\left[a_n\int_c^{c+T}\cos\left(\frac{2n\pi}{T}t\right)dt+b_n\int_c^{c+T}\sin\left(\frac{2n\pi}{T}t\right)dt\right]$$

$$=a_0T+0$$

$$\therefore a_0=\frac{1}{T}\int_c^{c+T}f(t)dt \quad --------(2)$$

Multiplying (1) by $\cos\left(\frac{2m\pi}{T}t\right)$ integrating, we get

$$\int_c^{c+T}f(t)\cos\left(\frac{2m\pi}{T}t\right)dt=$$

$$\int_c^{c+T}a_0\cos\left(\frac{2m\pi}{T}t\right)dt+$$

$$\sum_{n=1}^{\infty}\left(a_n\int_c^{c+T}\cos\left(\frac{2n\pi}{T}t\right)\cos\left(\frac{2m\pi}{T}t\right)dt+b_n\int_c^{c+T}\sin\left(\frac{2n\pi}{T}t\right)\cos\left(\frac{2m\pi}{T}t\right)dt\right)$$

$$=0+a_m\frac{T}{2}+0$$

$$\therefore a_n=\frac{2}{T}\int_c^{c+T}f(t)\cos\left(\frac{2n\pi}{T}t\right)dt \quad --------(3)$$

Multiplying (1) by $\sin\left(\frac{2m\pi}{T}t\right)$ integrating, we get

$$\int_c^{c+T} f(t)\sin\left(\frac{2m\pi}{T}t\right)dt = \int_c^{c+T} a_0\sin\left(\frac{2m\pi}{T}t\right)dt +$$

$$\sum_{n=1}^{\infty}\left(a_n\int_c^{c+T}\cos\left(\frac{2n\pi}{T}t\right)\sin\left(\frac{2m\pi}{T}t\right)dt + b_n\int_c^{c+T}\sin\left(\frac{2n\pi}{T}t\right)\sin\left(\frac{2m\pi}{T}t\right)dt\right)$$

$$= 0 + 0 + b_m\frac{T}{2}$$

$$\therefore b_n = \frac{2}{T}\int_c^{c+T} f(t)\sin\left(\frac{2n\pi}{T}t\right)dt \quad -------(4)$$

The Fourier coefficients a_0, a_n and b_n are given by the formulae (2), (3) and (4). These are called the Euler's formulae for Fourier coefficients.

Note:

(1) For $c = \frac{-T}{2}$, the formulae for the Fourier coefficients a_0, a_n and b_n becomes

$$a_0 = \frac{1}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}} f(t)dt = \begin{cases} 0, & \text{if } f(t) \text{ is odd} \\ \frac{2}{T}\int_0^{\frac{T}{2}} f(t)dt, & \text{if } f(t) \text{ is even} \end{cases}$$

$$a_n = \frac{2}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}} f(t)\cos\left(\frac{2n\pi}{T}t\right)dt = \begin{cases} 0, & \text{if } f(t) \text{ is odd} \\ \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\cos\left(\frac{2n\pi}{T}t\right)dt, & \text{if } f(t) \text{ is even} \end{cases}$$

$$b_n = \frac{2}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}} f(t)\sin\left(\frac{2n\pi}{T}t\right)dt = \begin{cases} \frac{4}{T}\int_0^{\frac{T}{2}} f(t)\sin\left(\frac{2n\pi}{T}t\right)dt, & \text{if } f(t) \text{ is odd} \\ 0, & \text{if } f(t) \text{ is even} \end{cases}.$$

Thus if $f(t)$ is an even function of t, then Fourier series expansion of $f(t)$ in $\left(\frac{-T}{2},\frac{T}{2}\right)$ is given by

$$f(t)=a_0+\sum_{n=1}^{\infty}a_n\cos\left(\frac{2n\pi}{T}t\right) \text{ where } a_0=\frac{2}{T}\int_0^{\frac{T}{2}}f(t)dt \text{ and } a_n=\frac{4}{T}\int_0^{\frac{T}{2}}f(t)\cos\left(\frac{2n\pi}{T}t\right)dt.$$

Such an expansion is called the Fourier Cosine series expansion of $f(t)$. Thus an even periodic function has Fourier Cosine series expansion.

If $f(t)$ is odd, then the Fourier series expansion of $f(t)$ in $\left(\frac{-T}{2},\frac{T}{2}\right)$ is given by

$$f(t)=\sum_{n=1}^{\infty}b_n\sin\left(\frac{2n\pi}{T}t\right) \text{ where } b_n=\frac{4}{T}\int_0^{\frac{T}{2}}f(t)\sin\left(\frac{2n\pi}{T}t\right)dt.$$

This expansion is called the Fourier Sine series expansion of $f(t)$. Note that an even periodic function has Fourier Sine series expansion.

2) For $T=2\pi$, the Fourier series expansion of $f(t)$ defined in $(c,c+2\pi)$ is given by

$$f(t)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos nt+b_n\sin nt\right)$$

where $$a_0=\frac{1}{2\pi}\int_c^{c+2\pi}f(t)dt,$$

$$a_n=\frac{1}{\pi}\int_c^{c+2\pi}f(t)\cos ntdt$$

and $$b_n=\frac{1}{\pi}\int_c^{c+2\pi}f(t)\sin ntdt.$$

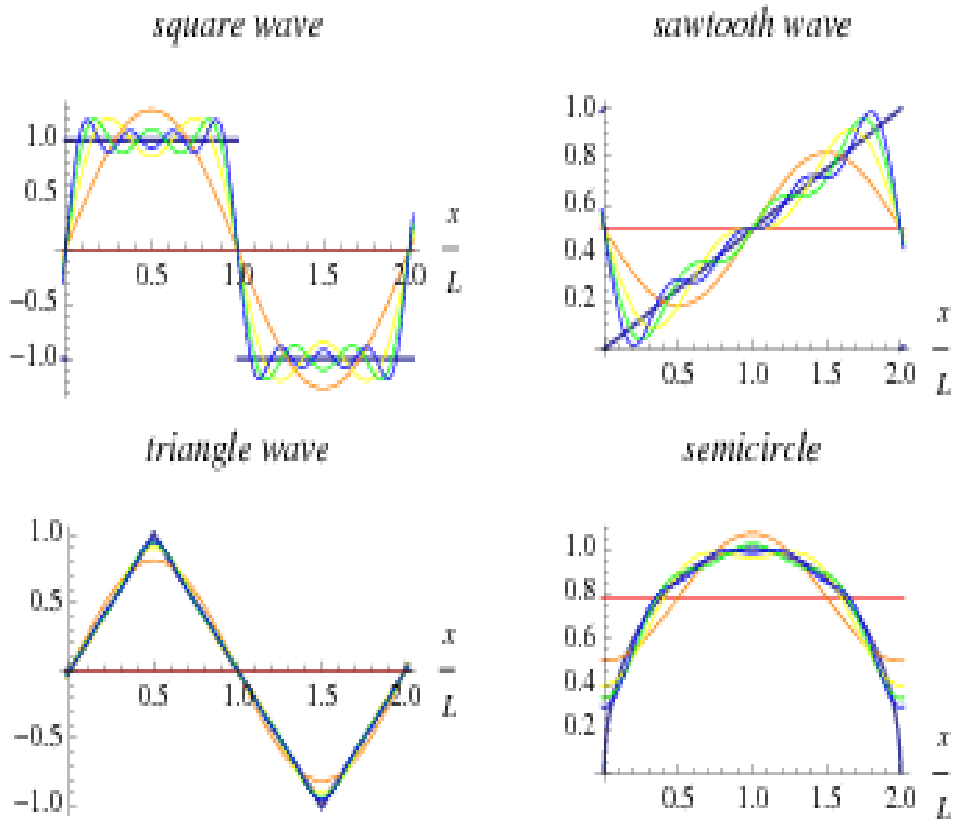
For $c=-\frac{\pi}{2}$ if f(t) is even function of t, then Fourier series expansion of f(t) is

$$f(t)=a_0+\sum_{n=1}^{\infty}a_n\cos nt \text{ where } a_0=\frac{1}{\pi}\int_0^{\pi}f(t)\,dt, a_n=\frac{2}{\pi}\int_0^{\pi}f(t)\cos nt\,dt.$$

If f (t) is odd function of t , then Fourier series expansion of f(t) is $f(t)=\sum_{n=1}^{\infty} b_n \sin nt$ where $b_n=\frac{2}{\pi}\int_0^{\pi} f(t)\sin nt\,dt.$

3) Let f(t) has Fourier series expansion $f(t)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos\frac{2n\pi t}{T}+b_n\sin\frac{2n\pi t}{T}\right)$

Consider the terms $a_n\cos\frac{2n\pi t}{T}+b_n\sin\frac{2n\pi t}{T}$ which is called the n[th] harmonic in the Fourier series expansion of f(t) for $a_n=r_n\cos\theta_n$, $b_n=r_n\sin\theta_n$ *or* $r_n=\sqrt{a_n^2+b_n^2}$ *and* $\theta_n=\tan^{-1}\left(\frac{b_n}{a_n}\right)$ the n[th] harmonic becomes $r_n\cos\theta_n\cos\frac{2n\pi t}{T}+r_n\sin\theta_n\sin\frac{2n\pi t}{T}=r_n\cos\left(\frac{2n\pi t}{T}-\theta_n\right)$. Then $r_n=\sqrt{a_n^2+b_n^2}$ is called the amplitude of the n[th] harmonic and $\theta_n=\tan^{-1}\left(\frac{b_n}{a_n}\right)$ is called phase angle of the n[th] harmonic.



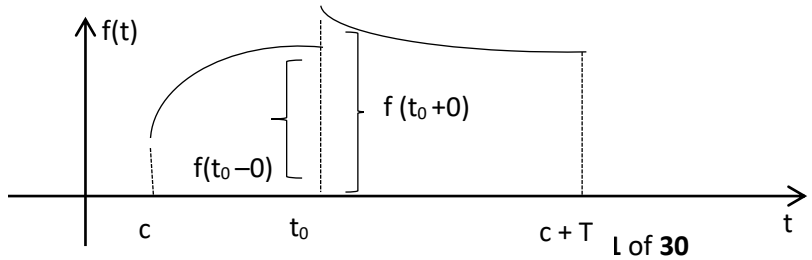
Dirichlet's condition for the convergence of the Fourier series

If a periodic function f (t) of period T, is piecewise continuous in the interval [c, c + T] and has left and right hand derivative at each point of that interval then

The Fourier series expansion $a_0+\sum_{n=1}^{\infty}\left(a_n\cos\frac{2n\pi t}{T}+b_n\sin\frac{2n\pi t}{T}\right)$ is convergent.

Its sum is f(t) except at a point t_0 at which f(t), is discontinuous and at t_0, it converges to the average of left and right hand limit of f(t) at t_0. i.e., to

$$\frac{1}{2}\left[f(t_0+0)+f(t_0-0)\right].$$



Parsevals identity:

If a periodic function f (t) has Fourier series expansion

$f(t)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos\frac{2n\pi t}{T}+b_n\sin\frac{2n\pi t}{T}\right)$ which is uniformly convergent in

[c, c + T] then $\frac{2}{T}\int_c^{c+T} f(t)\,dt=2a_0^2+\sum_{n=1}^{\infty}\left(a_n^2+b_n^2\right)$

Proof : Let f(t) has a Fourier series expansion

$f(t)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos\frac{2n\pi t}{T}+b_n\sin\frac{2n\pi t}{T}\right)$ which is uniformly convergent.

Consider $\int_c^{c+T}\left(f(t)\right)^2dt=\int_c^{c+T}\left(f(t)\right)\left[a_0+\sum_{n=1}^{\infty}\left(a_n\cos\frac{2n\pi t}{T}+b_n\sin\frac{2n\pi t}{T}\right)\right]dt$

$=a_0\int_c^{c+T} f(t)\,dt+\sum_{n=1}^{\infty}\left(a_n\int_c^{c+T} f(t)\cos\frac{2n\pi t}{T}dt+b_n\int_c^{c+T} f(t)\sin\frac{2n\pi t}{T}dt\right)$ since Fourier series is uniformly convergent

$=a_0\,T\,a_0+\sum_{n=1}^{\infty}\left(a_n\,\frac{T}{2}a_n+b_n\,\frac{T}{2}b_n\right)$

$=\frac{T}{2}\left(2{a_0}^2+\sum_{n=1}^{\infty}\left({a_n}^2+{b_n}^2\right)\right)$

Therefore, $\frac{2}{T}\int_c^{c+T} f(t)\,dt=2a_0^2+\sum_{n=1}^{\infty}\left(a_n^2+b_n^2\right)$ which is called Parseval's identity.

Problems:

1. Expand $f(x) = x - x^2,\ -\pi \le x < \pi$ f(x+2π)=f(x) , as a Fourier series.

Solution: Let $f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$. Then

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi}(x - x^2)\,dx = \frac{-2}{2\pi}\int_0^{\pi} x^2\,dx$$ since, x is odd and x^2 is even function.

$$= \frac{-1}{\pi}\left(\frac{x^3}{3}\right)_0^{\pi} = \frac{-\pi^3}{3\pi} = \frac{-\pi^2}{3}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\left(x - x^2\right)\cos nx\ dx = \frac{-2}{\pi}\int_0^{\pi} x^2 \cos nx\ dx$$

$$= \frac{-2}{\pi}\left(x^2\frac{\sin nx}{n} - 2x\left(\frac{-\cos nx}{n^2}\right) + 2\left(\frac{-\sin nx}{n^3}\right)\right)_0^{\pi}$$

$$= \frac{-2}{\pi}\left(\pi^2\frac{\sin n\pi}{n} - 0\frac{\sin n0}{n} + 2\pi\left(\frac{\cos n\pi}{n^2}\right) - 0\frac{\cos 0}{n^2} - \frac{2}{n^3}\left(\sin n\pi - \sin n0\right)\right)$$

$$= \frac{-2}{\pi}\left(0 - 0 + 2\pi\frac{(-1)^n}{n^2} - 0 - \frac{2}{n^3}.0\right)$$

$$= -\frac{4(-1)^n}{n^2} = \frac{4}{n^2}(-1)^{n+1}.$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\left(x - x^2\right)\sin nx dx = \frac{2}{\pi}\int_0^{\pi} x\sin nx dx$$

$$= \frac{2}{\pi}\left(x\left(\frac{-\cos nx}{n}\right) - 1\left(\frac{-\sin nx}{n^2}\right)\right)_0^{\pi}$$

$$= \frac{2}{\pi}\left(-\pi\frac{\cos n\pi}{n} - 0 + \frac{1}{n^2}\left(\sin n\pi - \sin no\right)\right)$$

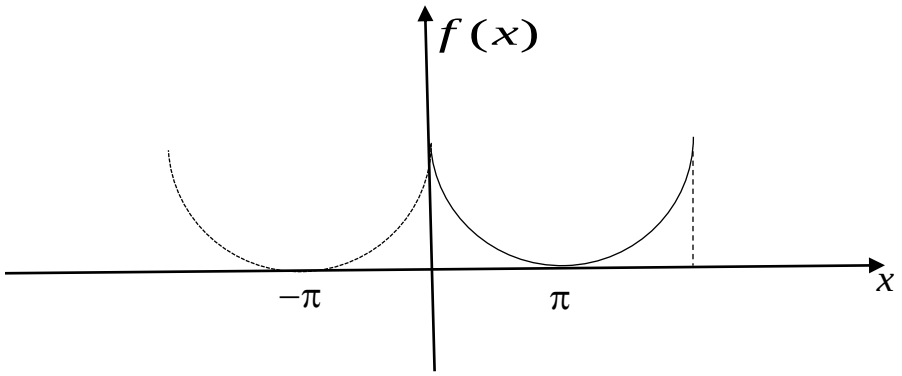
$$= \frac{-2}{n}\cos n\pi = \frac{-2}{n}(-1)^n = \frac{2}{n}(-1)^{n+1}.$$

$$\therefore f(x) = \frac{-\pi^2}{3} + \sum_{n=1}^{\infty}\left[\frac{4}{n^2}(-1)^{n+1}\cos nx + \frac{2}{n}(-1)^{n+1}\sin nx\right].$$

2. Obtain the Fourier series expansion of $f(x) = \frac{1}{4}(\pi - x)^2, \quad 0 \le x < 2\pi$

f(x+2π)=f(x) and hence obtain $(i)\sum\frac{1}{n^2}$ $(ii)\sum\frac{(-1)^{n+1}}{n^2}$ $(iii)\sum\frac{1}{(2n-1)^2}$

Solution:



Here $f(x)$ is an even function.

$$\therefore f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx, \quad \text{where}$$

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi}\frac{1}{4}(\pi - x)^2\,dx = \frac{1}{8\pi}\left[\frac{(\pi - x)^3}{3}\right]_0^{2\pi} = \frac{1}{24\pi}(\pi^3 + \pi^3) = \frac{\pi^2}{12}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi}\frac{1}{4}(\pi - x)^2\cos nx\,dx$$

$$= \frac{1}{4\pi}\left[(\pi - x)^2\frac{\sin nx}{n} + 2(\pi - x)\left(\frac{-\cos nx}{n^2}\right) + 2(-1)\left(\frac{-\sin nx}{n^3}\right)\right]_0^{2\pi}$$

$$= \frac{1}{4\pi}\left(0 - \frac{2}{n^2}(-\pi\cos 2n\pi - \pi\cos 0) + 0\right)$$

$$= \frac{-1}{4\pi}\frac{2}{n^2}(-2\pi) = \frac{1}{n^2}$$

$$\therefore f(x) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty}\frac{1}{n^2}\cos nx.$$

At x = 0, $f(0) = \frac{\pi^2}{4} = \frac{\pi^2}{12} + \sum_{n=1}^{\infty}\frac{1}{n^2}$

$$\therefore \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{4} - \frac{\pi^2}{12} = \frac{\pi^2}{6} \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

At $x = \pi,\ f(x) = 0$

$$0 = \frac{\pi^2}{12} + \sum_{n=1}^{\infty}\frac{1}{n^2}\cos n\pi$$

$$\therefore \sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} = \frac{-\pi^2}{12}$$

$$\therefore \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12} \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

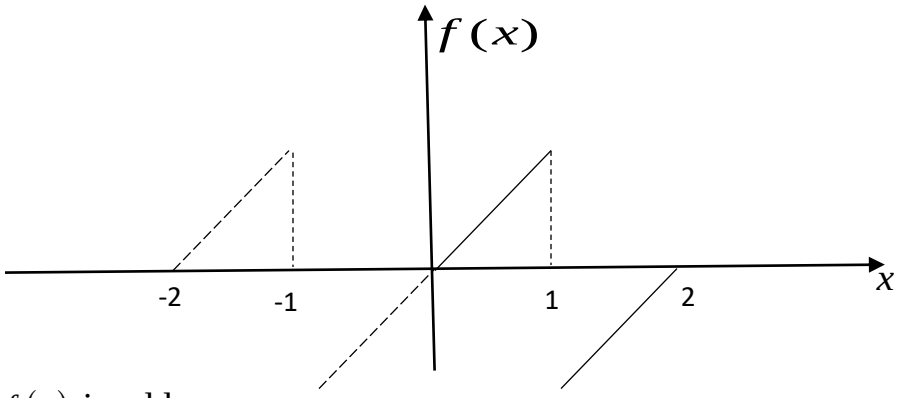
(1) + (2) gives

$$2\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi^2}{6} + \frac{\pi^2}{12} = \frac{\pi^2}{4}$$

$$\therefore \sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

3. Expand $f(x) = \begin{cases} \pi x, & 0 \le x < 1 \\ 0, & x = 1 \\ \pi(x-2), & 1 < x \le 2 \end{cases}$ $f(x+2) = f(x)$, as a Fourier series and hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots\ldots\ldots\ldots$

Solution:



$f(x)$ is odd,

$$\therefore f(x) = \sum_{n=1}^{\infty} b_n \sin n\pi x$$

Where $b_n = \frac{2}{2}\int_0^2 f(x)\sin n\pi x dx$

$$= \int_0^1 \pi x \sin \pi x dx + \int_1^2 \pi(x-2)\sin \pi x dx$$

$$= \pi\left[x\left(\frac{-\cos n\pi x}{n\pi}\right) - (1)\left(\frac{-\sin n\pi x}{n^2\pi^2}\right)\right]_0^1 + \left[\pi(x-2)\left(\frac{-\cos n\pi x}{n\pi}\right) - \left(\pi(1)\left(\frac{-\sin n\pi x}{n^2\pi^2}\right)\right)\right]_1^2$$

$$= \pi\left(\frac{-1}{n\pi}\cos n\pi - 0 + 0 + (0-0) - \frac{\pi((-1))(-\cos n\pi)}{n\pi}\right)$$

$$= \pi\left(\frac{-2\cos n\pi}{n\pi}\right) = \frac{-2}{n}(-1)^n = \frac{2}{n}(-1)^{n+1}$$

$$\therefore f(x) = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin n\pi x$$

$$f\left(\frac{1}{2}\right) = \frac{\pi}{2} = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi}{2}$$

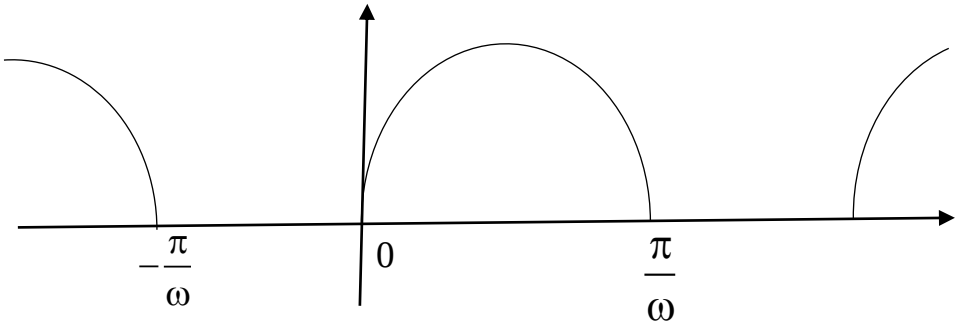
$$\left[\sin\frac{n\pi}{2} = \begin{cases} 0 & n \text{ is even} \\ (-1)^{\frac{n-1}{2}} & n \text{ is odd} \end{cases}\right]$$

$$\frac{\pi}{4} = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi}{2}$$

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{2n+1}=1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots .$$

4. Expand $f(t)=\begin{cases}0, & -\dfrac{\pi}{\omega}<t<0\\ E\sin\omega t, & 0<t<\dfrac{\pi}{\omega}\end{cases}$, $f\left(t+\dfrac{2\pi}{\omega}\right)=f(t),\forall t$ as a Fourier series.

Solution:



$f(t)$ is neither even nor odd.

$\therefore\ f(t)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos n\omega t+b_n\sin n\omega t\right)$ where

$$a_0=\frac{1}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}}f(t)dt=\frac{1}{\frac{2\pi}{\omega}}\int_0^{\frac{\pi}{\omega}}E\sin\omega t dt$$

$$=\frac{E\omega}{2\pi}\left(\frac{-\cos\omega t}{\omega}\right)_0^{\frac{\pi}{\omega}}=\frac{E}{\pi}.$$

$$a_n=\frac{2}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}}f(t)\cos n\omega t dt=\frac{E\omega}{2\pi}\int_0^{\frac{\pi}{\omega}}\left(\sin(1+n)\omega t+\sin(1-n)\omega t\right)dt$$

$$= \begin{cases} \dfrac{E\omega}{2\pi}\left(-\dfrac{\cos(1+n)\omega t}{(1+n)\omega} - \dfrac{\cos(1-n)\omega t}{(1-n)\omega} \right)_0^{\frac{\pi}{\omega}}, & n \neq 1 \\ \dfrac{E\omega}{2\pi}\int_0^{\frac{\pi}{\omega}} \sin 2\omega t dt, & n = 1 \end{cases}$$

$$= \begin{cases} \dfrac{E}{2\pi}\left(-\dfrac{\cos(1+n)\pi}{(1+n)} - \dfrac{\cos(1-n)\pi}{(1-n)} + \dfrac{1}{(1+n)} + \dfrac{1}{1-n} \right), & n \neq 1 \\ \dfrac{E\omega}{2\pi}\left(\dfrac{-\cos 2\omega t}{2\omega} \right)_0^{\frac{\pi}{\omega}}, & n = 1 \end{cases}$$

$$= \begin{cases} \dfrac{E}{2\pi}\left(\dfrac{2}{(1+n)} + \dfrac{2}{1-n} \right) = \dfrac{2E}{\left(n^2-1\right)\pi}, & n \text{ is even} \\ 0, & n \text{ is odd} \end{cases}.$$

$$b_n = \frac{2}{T}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(t)\sin wt\, dt = \frac{Ew}{\pi}\int_0^{\frac{\pi}{W}} \sin wt\, \sin nwt\, dt$$

$$= \frac{Ew}{2\pi}\int_0^{\frac{\pi}{W}} \cos(n-1)wt\, - \cos(n+1)wt\, dt$$

$$= \frac{Ew}{2\pi}\left(\frac{\sin(n-1)wt}{(n-1)w} - \frac{\sin(n+1)wt}{(n+1)w} \right)_0^{\frac{\pi}{w}} \quad n \neq 1$$

$$= 0, \; n \neq 1$$

$$b_1 = \int_0^{\frac{\pi}{w}} \frac{Ew}{\pi} \sin^2 wt\, dt$$

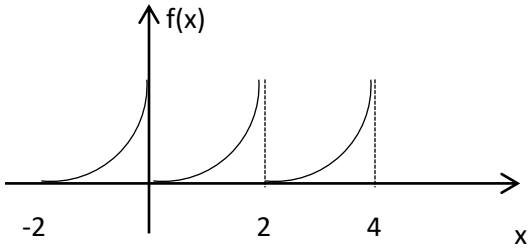
$$= \frac{Ew}{2\pi} \int_0^{\frac{\pi}{W}} (1 - \cos 2wt)\, dt$$

$$= \frac{Ew}{2\pi} \left(t - \frac{\sin 2wt}{2w} \right)_0^{\frac{\pi}{w}}$$

$$= \frac{Ew}{2\pi} \left(\frac{\pi}{w} - 0 \right) = \frac{E}{2}$$

$$\therefore f(t) = \frac{E}{\pi} + \frac{E}{2} \sin wt - \frac{2E}{\pi} \left(\frac{\cos 2wt}{2^2 - 1} + \frac{\cos 4wt}{4^2 - 1} + ... \right)$$

5. Expand $f(x) = x^2, 0 \le x \le 2, \quad f(x+2) = f(x)$ as a Fourier series and hence evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2}$.



Solution: Note that f(x) is neither even nor odd.

$$\therefore f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where

$$a_0 = \frac{1}{2} \int_0^2 x^2\, dx = \frac{1}{2} \left(\frac{x^3}{3} \right)_0^2 = \frac{4}{3}$$

$$a_n = \frac{2}{2} \int_0^2 x^2 \cos n\pi x\, dx = \left(\frac{x^2 \sin n\pi x}{n\pi} - 2x \left(\frac{-\cos n\pi x}{n^2 \pi^2} \right) + 2 \left(\frac{-\sin n\pi x}{n^3 \pi^3} \right) \right)_0^2 = \frac{4}{n^2 \pi^2}$$

$$b_n = \frac{2}{2}\int_0^2 x^2 \sin n\pi x\, dx = \left(\frac{-x^2 \cos n\pi x}{n\pi} - 2x\left(\frac{-\sin n\pi x}{n^2\pi^2} \right) + 2\left(\frac{\cos n\pi x}{n^3\pi^3} \right) \right)_0^2$$

$$= \frac{-4}{n\pi}\cos 2n\pi - 0 - 0 + \frac{2}{n^3\pi^3}(\cos 2n\pi - \cos 0)$$

$$= \frac{-4}{n\pi}.$$

$$\therefore f(x) = \frac{4}{3} + \sum_{n=1}^{\infty}\left(\frac{4}{n^2\pi^2}\cos n\pi x - \frac{4}{n\pi}\sin n\pi x \right)$$

A x=0, f(x) is discontinuous

$$\frac{f(0+) + f(0-)}{2} = \frac{4}{3} + \sum_{n=1}^{\infty}\left(\frac{4}{n^2\pi^2}\cos 0 - \frac{4}{n\pi}\sin 0 \right)$$

$$\Rightarrow \frac{4+0}{2} = \frac{4}{3} + \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}$$

$$\therefore \sum_{n=1}^{\infty}\frac{1}{n^2} = \left(2 - \frac{4}{3}\right)\frac{\pi^2}{4} = \frac{\pi^2}{6}$$

Exercises : Expand the following functions as Fourier series

1. $f(x)=x, 0\le x\le 2\pi,\ f(x+2\pi)=f(x)$

2. $f(x)=e^{-x}, 0<x\le 2\pi,\ f(x+2\pi)=f(x)$ and hence deduce that $\dfrac{\pi}{2\sinh\pi}=\sum_{n=1}^{\infty}\dfrac{(-1)^n}{n^2+1}$

3. $f(x)=x\sin x, 0<x<2\pi,\ f(x+2\pi)=f(x)$ and hence deduce that $\sum_{n=1}^{\infty}\dfrac{1}{n^2-1}=\dfrac{3}{4}$

4. $f(x)=\begin{cases} x & 0\le x\le \pi \\ 2\pi-x & \pi<x<2\pi \end{cases}$ $f(x+2\pi)=f(x)$

5. $f(x)=|\sin x|, 0\le x\le 2\pi,\ f(x+2\pi)=f(x)$

6. $f(x)=\sin \pi x, 0\le x<1,\ f(x+1)=f(x)$

7. $f(x)=2x-x^2, 0\le x\le 2,\ f(x+2)=f(x)$

8. $f(x)=2-|x|\ \ -2\le x<2,\ f(x+4)=f(x)$

9. $f(t)=\begin{cases} 1-\dfrac{|t|}{a} & |t|\le a \\ 0 & a<|t|<T/2 \end{cases}$ $f(t+T)=f(t)$

10. $f(x)=\begin{cases} x & \dfrac{-\pi}{2}\le x\le \dfrac{\pi}{2} \\ \pi-x & \dfrac{\pi}{2}<x<3\pi/2 \end{cases}$ $f(x+2\pi)=f(x)$

Half range Expansions

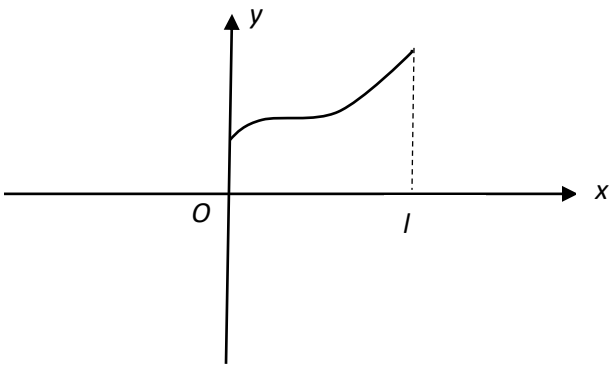
While solving various physical and engineering problems, there is a practical need for expanding functions defined over a finite range. Such an expansion is possible if functions under consideration can be extended to a periodic function which is either even or odd.

Consider a piecewise continuous function $f(x)$, defined in a finite interval $(0,\ l)$. Then it is possible to extend $f(x)$ to a periodic function, which is even or odd.

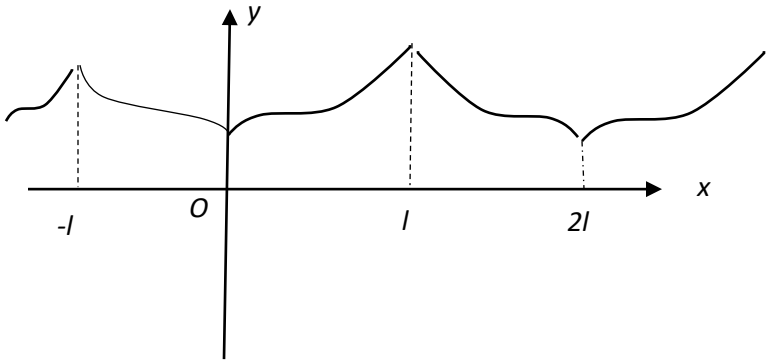
Consider the function $g(x)$ defined as follows:

$$g(x)=\begin{cases} f(x), & 0\le x\le l \\ f(-x), & -l<x<0 \end{cases} \quad ; \quad g(x+2l)=g(x).$$

Then $g(x)$ is called an even periodic extension of $f(x)$.



Graph of $f(x)$



Graph of even periodic extension of f(x).

The function $g(x)$ can be expanded as Fourier cosine series

$$g(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi}{l}x$$

where $a_0 = \frac{1}{l}\int_0^l g(x)\,dx$ and $a_n = \frac{2}{l}\int_0^l g(x)\,\cos\frac{n\pi}{l}x\,dx$

But for $0 \le x \le l,\ g(x) = f(x)$. We have

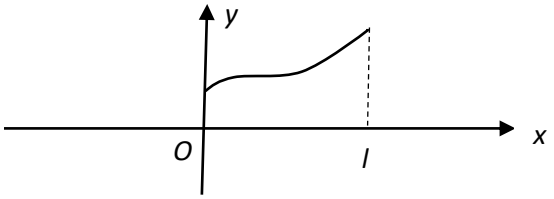
$$a_0 = \frac{1}{l}\int_0^l f(x)\,dx \text{ and } a_n = \frac{2}{l}\int_0^l f(x)\,\cos\frac{n\pi}{l}x\,dx$$

$$f(x) = g(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi}{l}x, \text{ for } 0 \le x \le l.$$

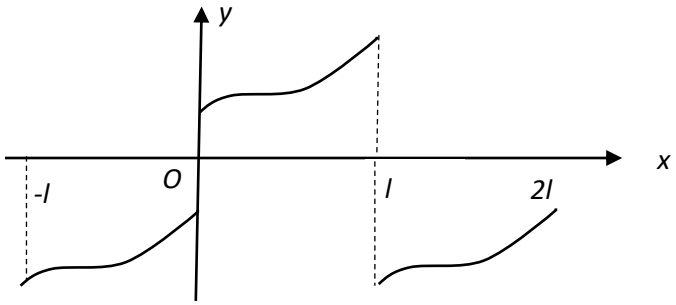
Such an expansion of $f(x)$ is called the <u>half range Fourier Cosine series expansion</u> of $f(x)$.

Also, if $g(x) = \begin{cases} f(x), & 0 \le x \le l \\ -f(-x), & -l < x < 0 \end{cases}$; $g(x+2l) = g(x)$.

Then $g(x)$ is called an odd periodic extension of $f(x)$.



Graph of $f(x)$



Graph of odd periodic extension of f(x)

The function $g(x)$ can be expanded as Fourier Sine series

$$g(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{l} x$$

where $b_n = \frac{2}{l} \int_0^l g(x) \sin \frac{n\pi}{l} x \, dx$

But for $0 \le x \le l,\ g(x) = f(x).$ We have

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi}{l} x \, dx$$

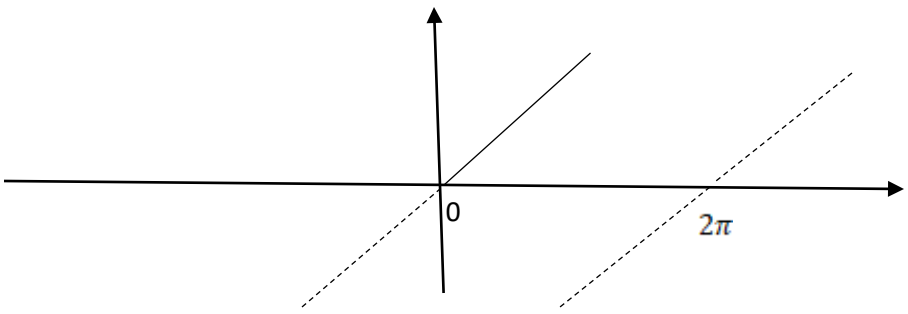
$$f(x) = g(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{l} x, \text{ for } 0 \le x \le l.$$

Which is called the half range Fourier Sine series expansion of $f(x).$

Problems:

1. Expand $f(x) = x, 0 \le x \le \pi$ as half range Fourier cosine and sine series. Also draw the graph of the corresponding periodic extensions of $f(x).$

   Solution:



Graph of even periodic extension of $f(x)$.

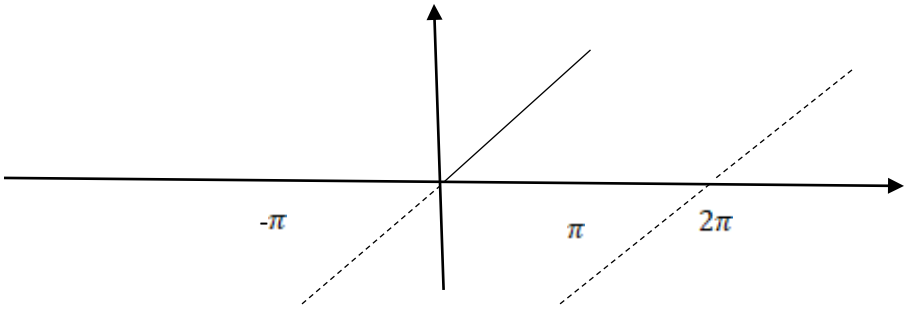
$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$, where

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)dx = \frac{1}{\pi}\int_0^{\pi} xdx = \frac{\pi}{2}.$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nxdx = \frac{2}{\pi}\int_0^{\pi} x\cos nxdx = \frac{2}{\pi}\left(\frac{x\sin nx}{n} - (1)\frac{(-\cos nx)}{n^2}\right)_0^{\pi}$$

$$= \frac{2}{\pi}\left(\frac{(-1)^n - 1}{n^2}\right) = \begin{cases} 0, & n \text{ even} \\ \frac{-4}{\pi n^2}, & n \text{ odd} \end{cases}.$$

$$\therefore f(x) = x = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \cdots\right).$$



Graph of odd periodic extension of $f(x)$.

$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$, where

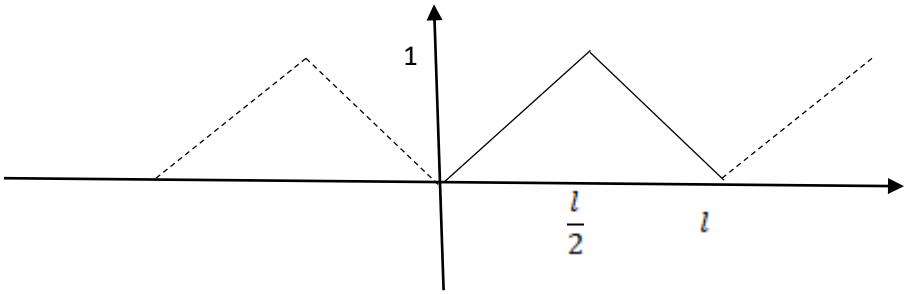
$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nxdx = \frac{2}{\pi}\int_0^{\pi} x\sin nxdx = \frac{2}{\pi}\left(\frac{x(-\cos nx)}{n} - (1)\frac{(-\sin nx)}{n^2}\right)_0^{\pi}$$

$$= \frac{2}{\pi}\left(\frac{-\pi(-1)^n + 0}{n}\right) = \frac{2}{n}(-1)^{n+1}.$$

$$\therefore f(x) = x = \sum_{n=1}^{\infty} \frac{2}{n}(-1)^{n+1}\sin nx.$$

2. Expand $f(x)=\begin{cases} \dfrac{x}{l}, & 0\le x\le \dfrac{l}{2} \\ 1-\dfrac{x}{l}, & \dfrac{l}{2}\le x\le l \end{cases}$ as half range Fourier cosine and sine series. Also draw the graph of the corresponding periodic extensions of $f(x)$.

Solution:



Graph of even periodic extension of $f(x)$.

$$f(x)=a_0+\sum_{n=1}^{\infty}a_n\cos\frac{n\pi}{l}x$$

Where $a_0=\frac{1}{l}\int_0^l f(x)\,dx=\frac{1}{l}\left[\int_0^{\frac{l}{2}}\frac{x}{l}\,dx+\int_{\frac{l}{2}}^{l}\left(1-\frac{x}{l}\right)dx\right]$

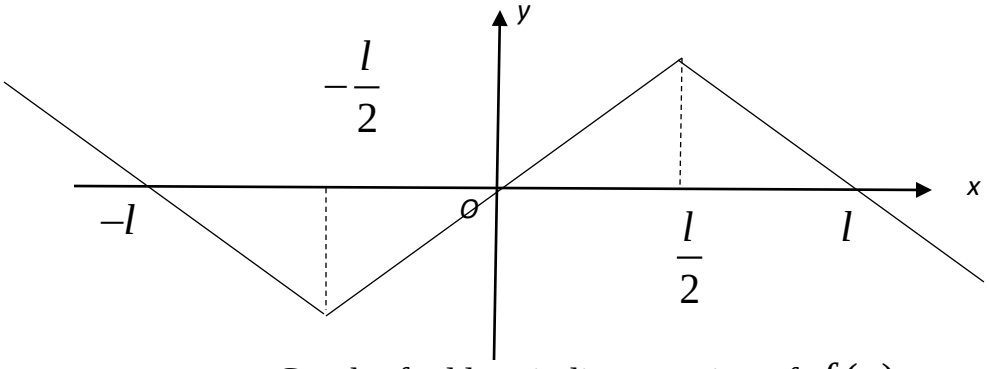
$$=\left[\left(\frac{x^2}{2l}\right)_0^{\frac{l}{2}}+\left(-\frac{1}{2l}\right)\left(\left(1-\frac{x}{l}\right)^2\right)_{\frac{l}{2}}^{l}\right]=\frac{1}{4}$$

$$a_n=\frac{2}{l}\int_0^l f(x)\cos\frac{n\pi}{l}x\,dx=\frac{2}{l}\left[\int_0^{\frac{l}{2}}\frac{x}{l}\cos\frac{n\pi}{l}x\,dx+\int_{\frac{l}{2}}^{l}\left(1-\frac{x}{l}\right)\cos\frac{n\pi}{l}x\,dx\right]$$

$$= \frac{2}{n^2\pi^2}\left(2\cos\frac{n\pi}{2} - \cos n\pi - 1\right)$$

$$= \begin{cases} 0, \ n \neq 2, \ 6, \ 10, \ 14, \ \ldots \\ -\dfrac{8}{n^2\pi^2}, \ \ n \neq 2, \ 6, \ 10, \ 14, \ \ldots \end{cases}$$

$$f(x) = \frac{1}{4} - \frac{8}{\pi^2}\left(\frac{1}{2^2}\cos\frac{2\pi}{l}x + \frac{1}{6^2}\cos\frac{6\pi}{l}x + \cdots\right).$$



Graph of odd periodic extension of $f(x)$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{l}x$$

Where $a_n = \dfrac{2}{l}\int_0^l f(x)\sin\dfrac{n\pi}{l}x\,dx$

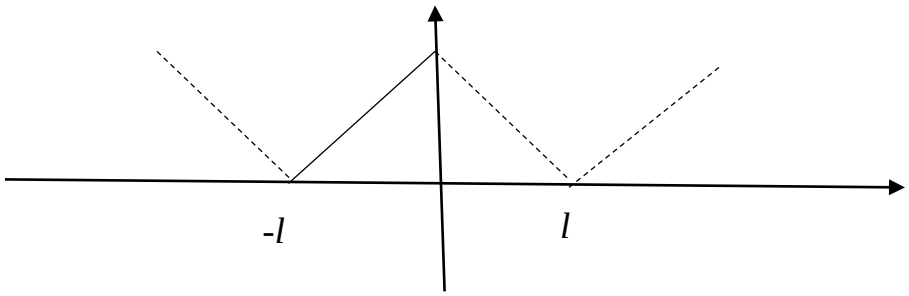
$$= \frac{2}{l}\left[\int_0^{\frac{l}{2}} \frac{x}{l} \ \sin\frac{n\pi}{l}x\,dx + \int_{\frac{l}{2}}^{l}\left(1 - \frac{x}{l}\right)\sin\frac{n\pi}{l}x\,dx\right]$$

$$= \frac{4}{n^2\pi^2}\sin\frac{n\pi}{2}x = \begin{cases} 0, & n \text{ even} \\ \frac{4}{n^2\pi^2}(-1)^{\frac{n-1}{2}}, & n \text{ odd} \end{cases}$$

$$\therefore \quad f(x) = \frac{4}{\pi^2}\left(\frac{1}{1^2}\sin\frac{\pi}{l}x - \frac{1}{3^2}\sin\frac{3\pi}{l}x + \frac{1}{5^2}\sin\frac{5\pi}{l}x - \cdots\right)$$

Expand $f(x) = 1 - \frac{x}{l}, 0 < x < l$ as a Fourier cosine and sine series. Also draw the graph of the corresponding periodic extensions of $f(x)$.

Solution:



Graph of even periodic extension of $f(x)$.

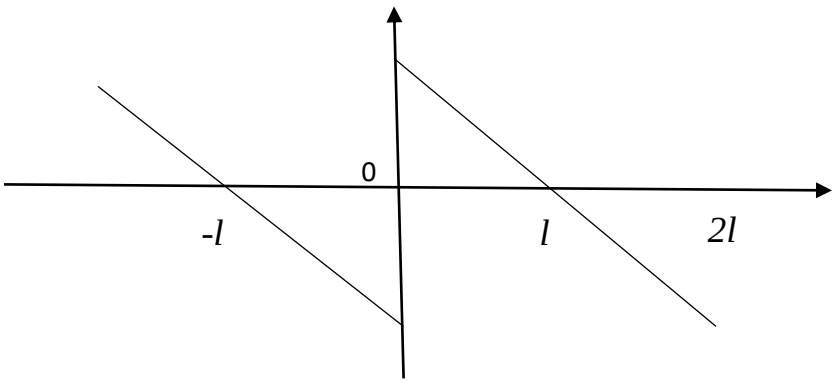
Let $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi}{l}x$.

Then $a_0 = \frac{2}{l}\int_0^l f(x)dx = \frac{1}{2}$.

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi}{l}x\,dx = \frac{2}{l}\int_0^l\left(1-\frac{x}{l}\right)\cos\frac{n\pi}{l}x\,dx$$

$$= \frac{2}{n^2\pi^2}\left(1-(-1)^n\right) = \begin{cases} 0, & n \text{ even} \\ \dfrac{4}{n^2\pi^2}, & n \text{ odd} \end{cases}.$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\frac{4}{\pi^2(2n-1)^2}\cos\frac{(2n-1)\pi x}{l}.$$



Graph of odd periodic extension of $f(x)$

Let $f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{l}x.$

Then $b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi}{l}x\,dx = \frac{2}{l}\int_0^l (1-\frac{x}{l})\sin\frac{n\pi}{l}x\,dx = \frac{2}{n\pi}$

Thus $f(x) = \sum_{n=1}^{\infty}\frac{2}{n\pi}\sin\frac{n\pi}{l}x.$

Problems:

Obtain the half range Fourier Cosine and sine series expansions of the functions. Also draw the graph of corresponding periodic extensions.

1) $f(x) = x\sin x, 0 < x < \pi.$

2) $f(x) = x(\pi - x), 0 < x < \pi.$

3) $f(x)=\begin{cases} x, & 0\le x\le \dfrac{\pi}{2} \\ \pi - x, & \dfrac{\pi}{2} < x < \pi \end{cases}$.

4) $f(x)=2-x, 0<x<2$.

5) $f(x)=\cos\dfrac{\pi}{l}x, 0<x<l$.

6) $f(x)=\sin\dfrac{\pi}{l}x, 0<x<l$.

7) If $f(x)$ is piecewise continuous for $0<x<l,$ having half range expansions

$f(x)=a_0+\sum_{n=1}^{\infty}a_n\cos\dfrac{n\pi}{l}x$ & $f(x)=\sum_{n=1}^{\infty}b_n\sin\dfrac{n\pi}{l}x$ then show that

$\dfrac{2}{l}\int_0^l (f(x))^2\,dx=2a_0{}^2+\sum_{n=1}^{\infty}a_n{}^2$ & $\dfrac{2}{l}\int_0^l (f(x))^2\,dx=\sum_{n=1}^{\infty}b_n{}^2$.